Research Question

Do improvements in renewable energy usage, forest coverage, and air pollution levels correlate with changes in citizens' well-being — measured by happiness scores and the prevalence of anxiety and depressive disorders — across countries over time, and do these relationships hold when controlling for economic development?

Motivation

As climate change accelerates and mental health crises grow globally, understanding the intersection between environmental quality and psychological well-being has become increasingly urgent. While policymakers often justify green energy transitions on ecological grounds, their potential impact on public mental health remains underexplored at the national level. This project aims to provide empirical evidence on whether countries that invest in cleaner energy and better environmental conditions also enjoy measurably higher levels of citizen well-being.

Data Sources

The primary dataset is the World Happiness Report (2015–2022) by Helliwell et al., Oxford Wellbeing Research Centre / Gallup, which provides annual happiness scores and GDP per capita. Environmental indicators are drawn from the Global Data on Sustainable Energy (IEA / World Bank) for renewable energy usage, the World Bank Open Data portal for forest area and PM2.5 air pollution concentrations. Mental health outcomes — anxiety and depressive disorder prevalence — are sourced from the IHME Global Burden of Disease Study 2023 via Our World in Data.

Data Collection and Characteristics

All datasets are downloaded in CSV format and merged by country and year using Python, producing a panel dataset of approximately 780 observations (130 countries × 6 years, 2015–2020). Countries missing more than 20% of variables will be excluded. Forest area and PM2.5 are used as complementary proxies for environmental quality, as a global urban green space index is not publicly available.

Planned Analysis

Analysis will proceed from exploratory data analysis and correlation tests to multiple regression, with fixed effects panel regression applied if feasible to isolate within-country changes over time. All work will be conducted in Python (Jupyter Notebook).